

Course code:ITA0608

Course Name:Machine Learning

Experiment 3 :Demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.

AIM : To implement and demonstrate the Decision Tree based ID3 algorithm using the Play Tennis dataset and classify a new sample based on the generated decision tree.

ALGORITHM:

1. Start.
2. Load the Play Tennis dataset.
3. Calculate the entropy of the target attribute.
4. Compute the information gain for each attribute.
5. Select the attribute with the highest information gain as the root node.
6. Split the dataset based on the selected attribute.
7. Repeat Steps 3–6 recursively for each subset until:
 - All samples belong to the same class, or
 - No attributes remain for further splitting.
8. Construct the Decision Tree.
9. Provide a new sample as input.
10. Traverse the Decision Tree using the attribute values of the new sample.
11. Predict the class label (Yes/No) for the new sample.
12. Stop.

CODE :

```
import pandas as pd
import math
data = pd.read_csv("play_tennis.csv")
def entropy(target):
    values = target.value_counts()
    e = 0
    total = len(target)
    for v in values:
        p = v / total
        e -= p * math.log2(p)
    return e
def information_gain(data, attribute, target):
    total_entropy = entropy(data[target])
    values = data[attribute].unique()
    weighted_entropy = 0
    for value in values:
        subset = data[data[attribute] == value]
        weighted_entropy += (len(subset) / len(data)) * entropy(subset[target])
    return total_entropy - weighted_entropy
def id3(data, original_data, features, target, parent_node=None):
    if len(data[target].unique()) == 1:
        return data[target].iloc[0]
    elif len(data) == 0:
        return original_data[target].mode()[0]
    elif len(features) == 0:
        return parent_node
    parent_node = data[target].mode()[0]
    gains = [information_gain(data, feature, target) for feature in features]
```

OUTPUT:

```
... Decision Tree:
{'Outlook': {'Sunny': {'Humidity': {'High': 'No', 'Normal': 'Yes'}}}, 'Overcas

New Sample:
{'Outlook': 'Sunny', 'Temperature': 'Cool', 'Humidity': 'High', 'Wind': 'Stro
Prediction: No
```

RESULT:

The Decision Tree based ID3 algorithm was successfully implemented using the Play Tennis dataset. The decision tree was constructed based on information gain, and the given new sample was correctly classified using the generated decision tree.